

BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

Complete

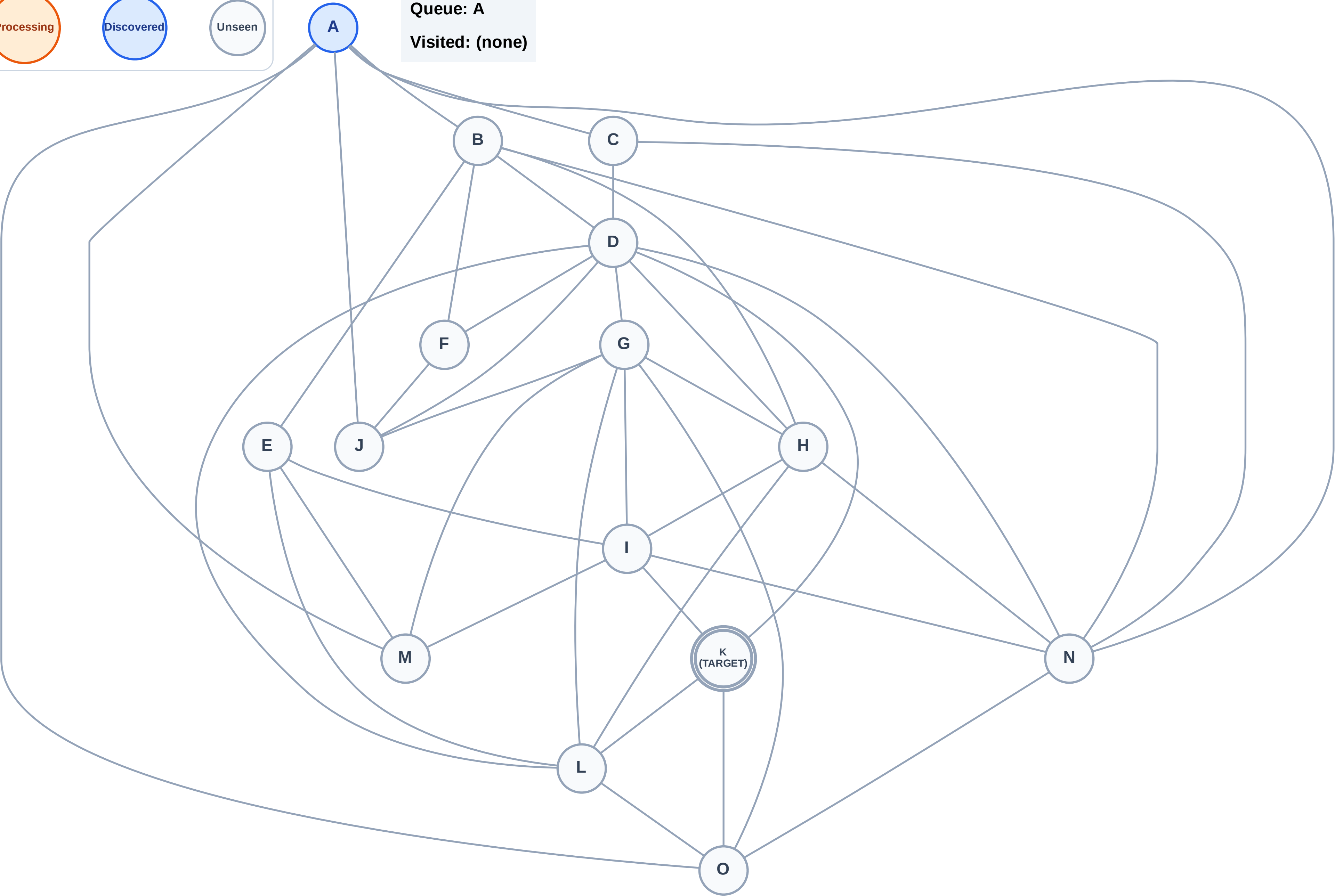
Processing

Discovered

Unseen

Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

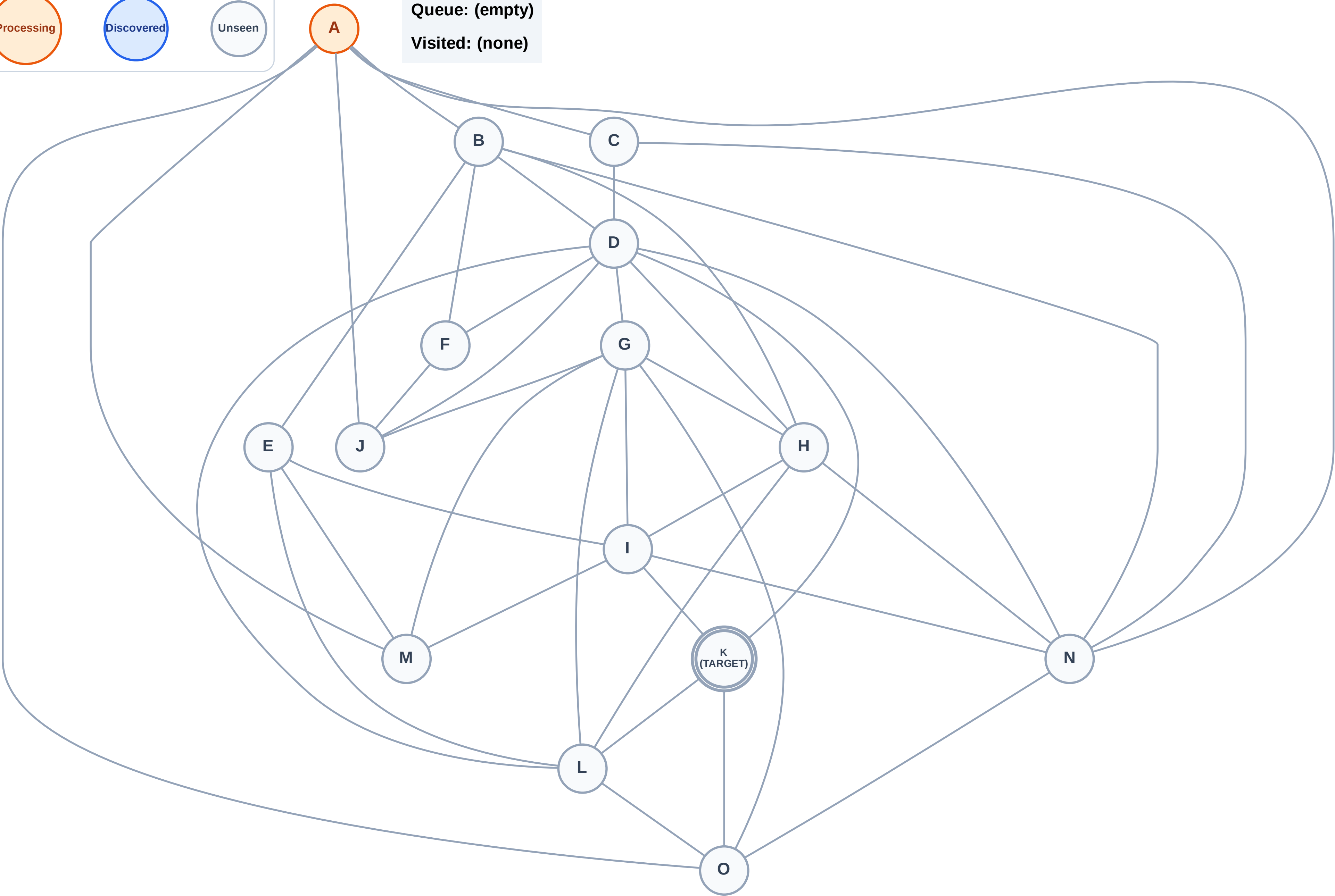
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: (none)



BFS Traversal

Step 3: Discover B, C, J, M, N, O from A

Legend

Complete

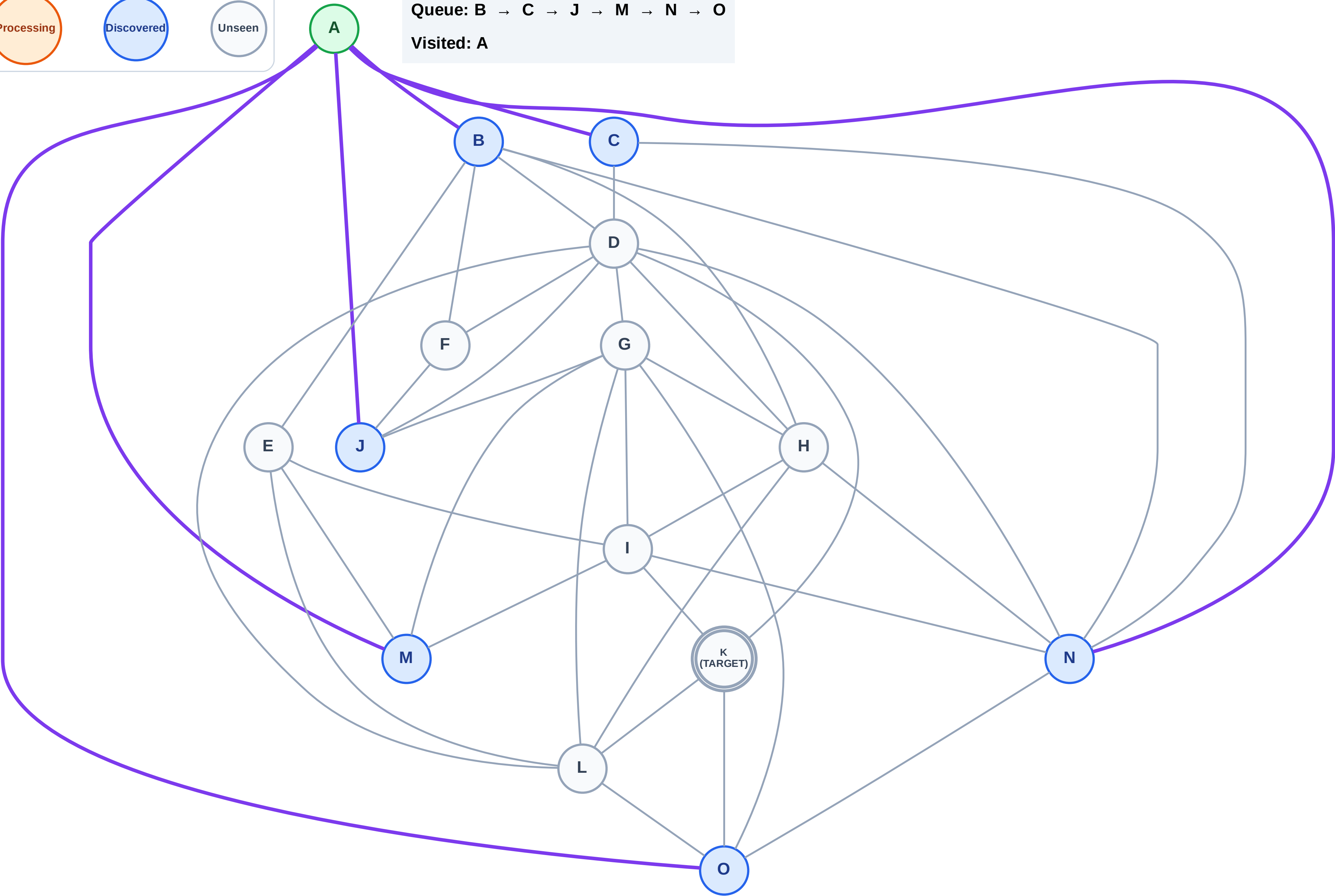
Processing

Discovered

Unseen

Queue: B → C → J → M → N → O

Visited: A



BFS Traversal

Step 4: Process node B

Legend

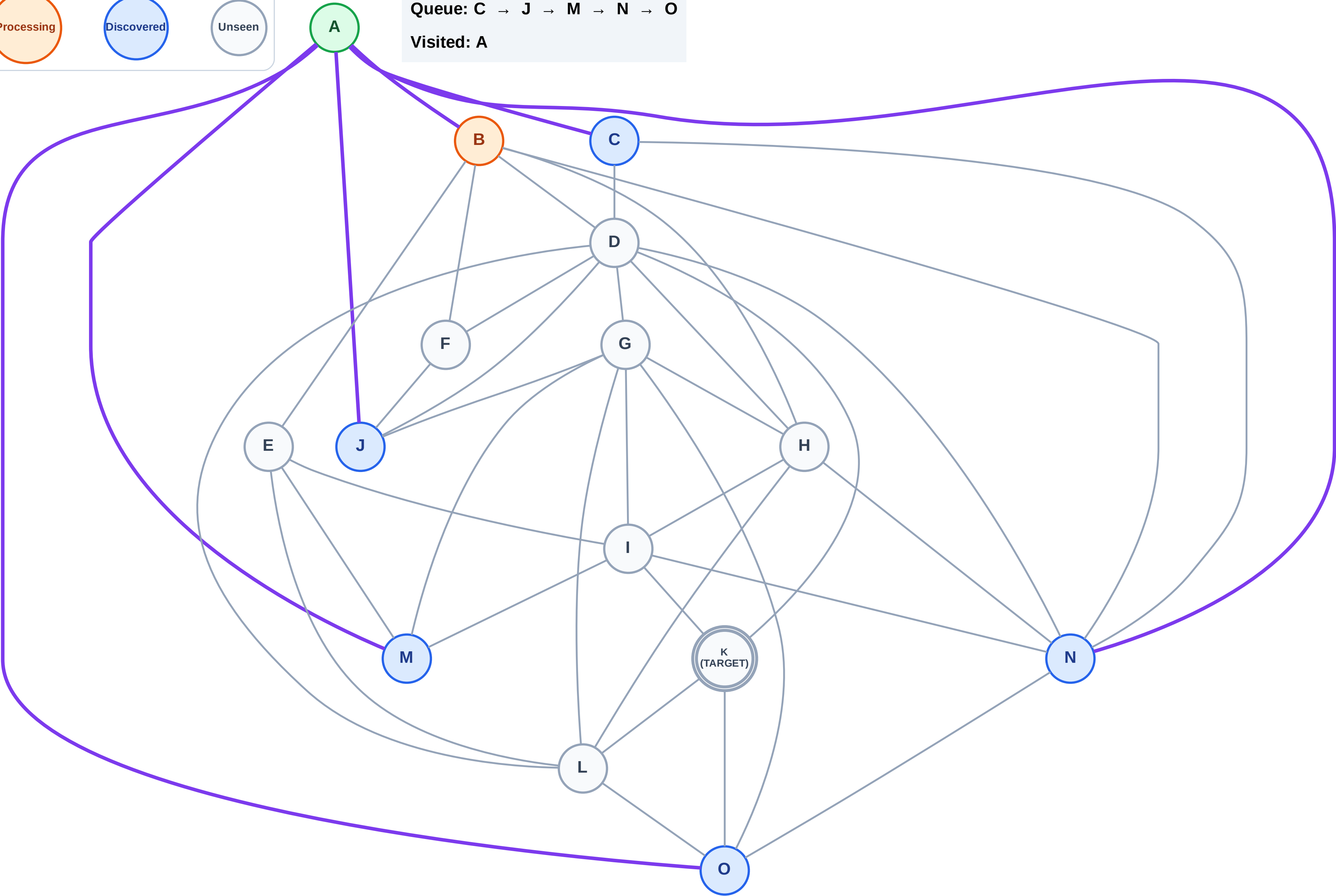
Complete

Processing

Discovered

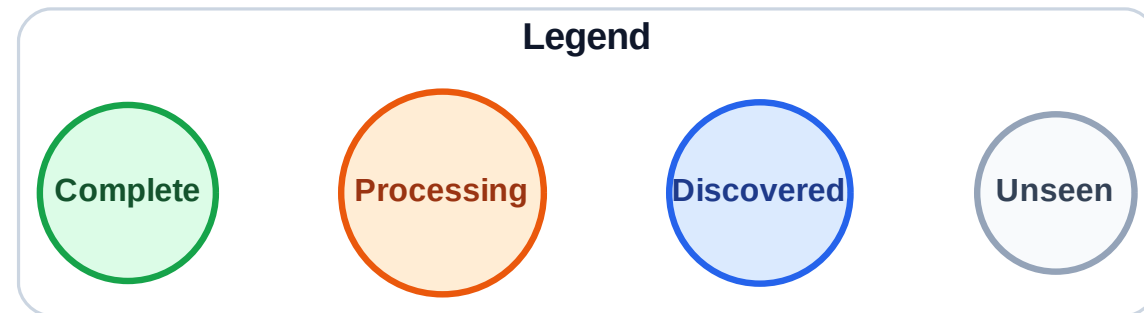
Unseen

Queue: C → J → M → N → O
Visited: A

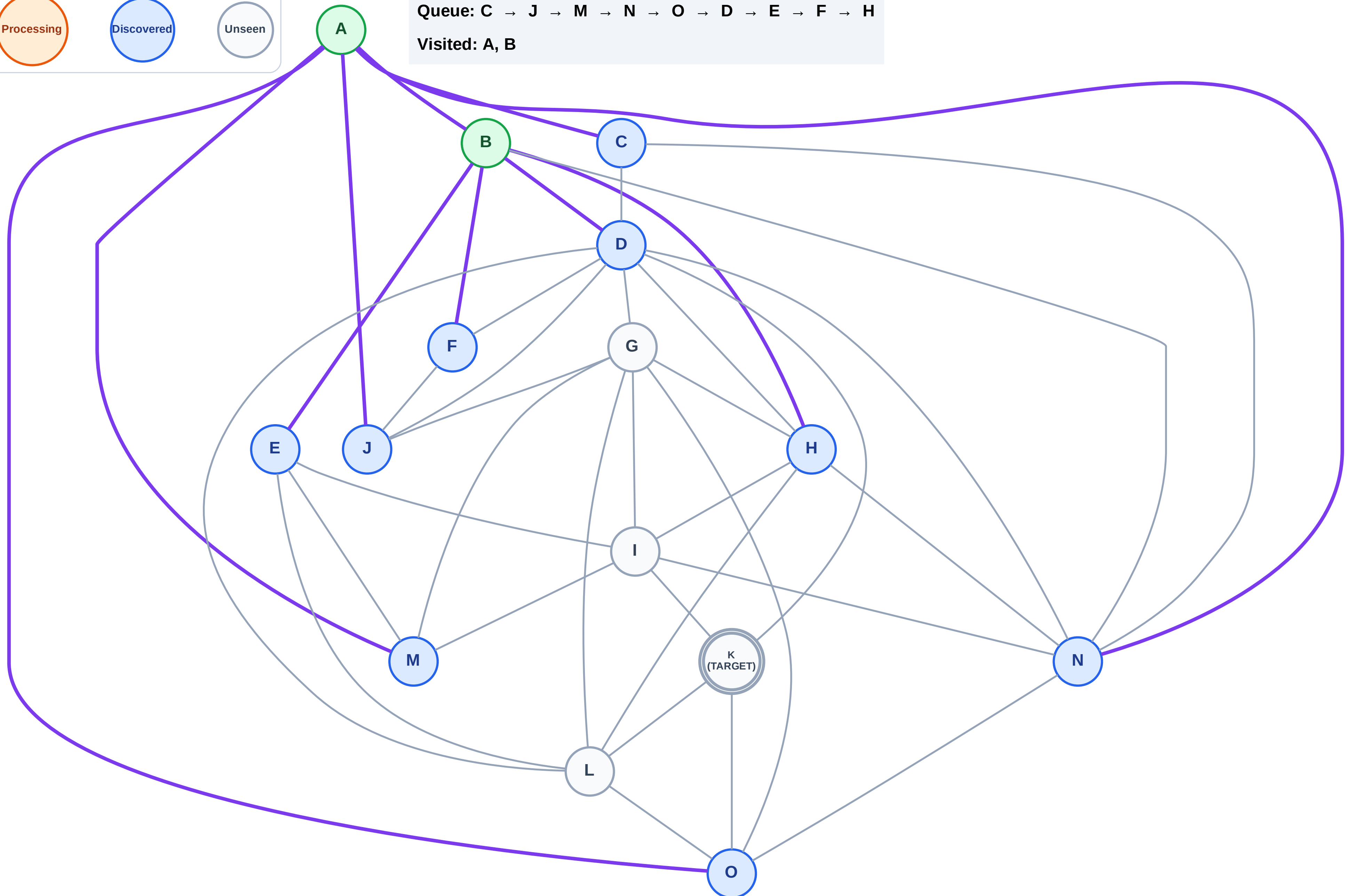


BFS Traversal

Step 5: Discover D, E, F, H from B



Queue: C → J → M → N → O → D → E → F → H
Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

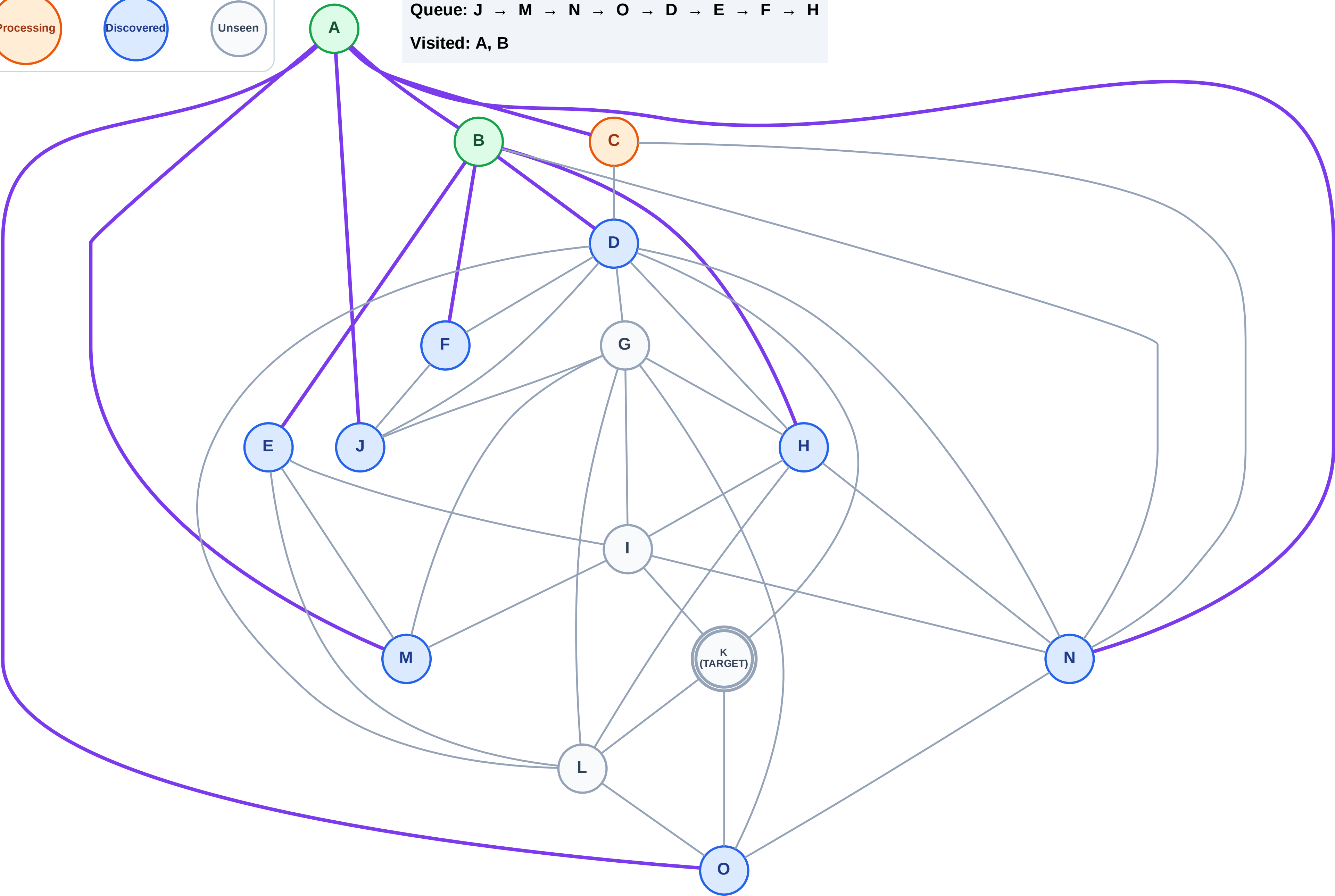
Processing

Discovered

Unseen

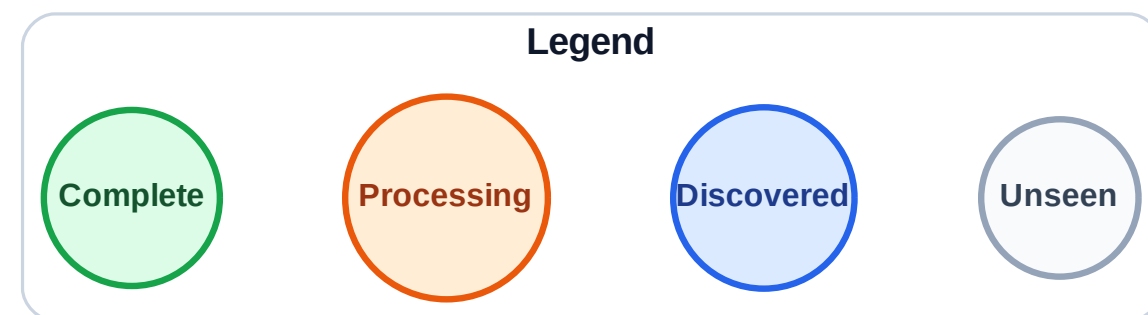
Queue: J → M → N → O → D → E → F → H

Visited: A, B

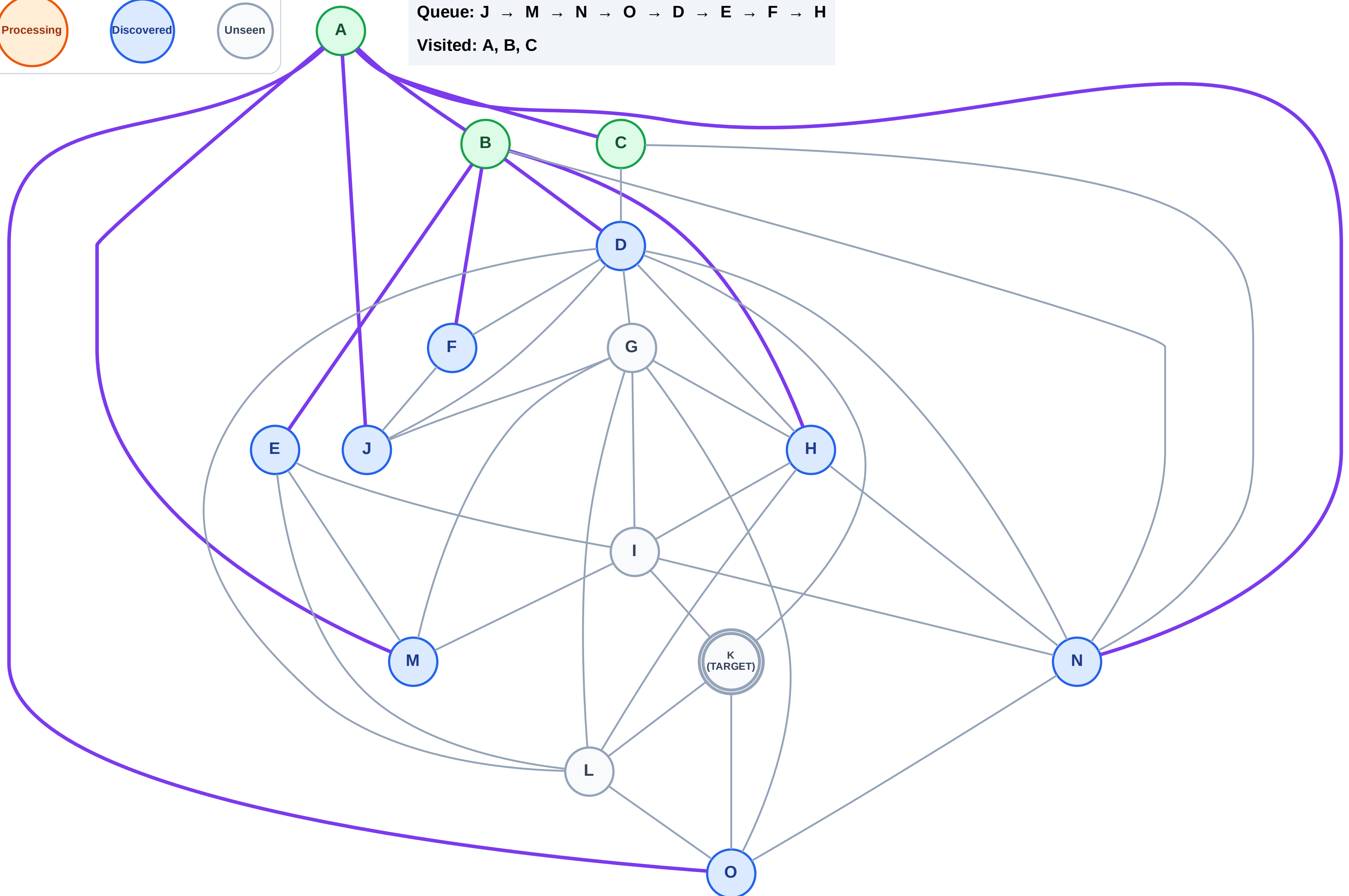


BFS Traversal

Step 7: No new neighbors from C



Queue: J → M → N → O → D → E → F → H
Visited: A, B, C



BFS Traversal

Step 8: Process node J

Legend

Complete

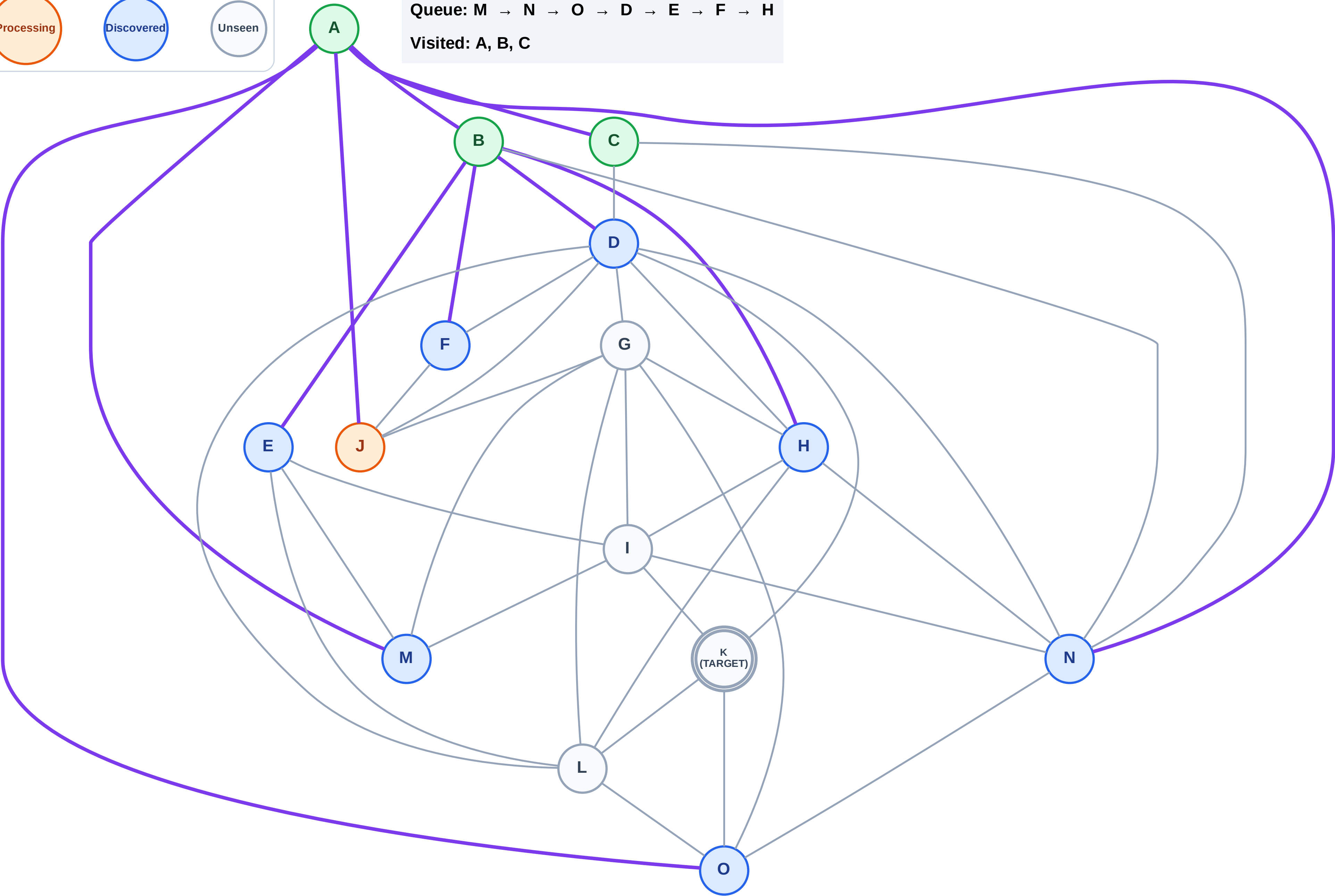
Processing

Discovered

Unseen

Queue: M → N → O → D → E → F → H

Visited: A, B, C



BFS Traversal

Step 9: Discover G from J

Legend

Complete

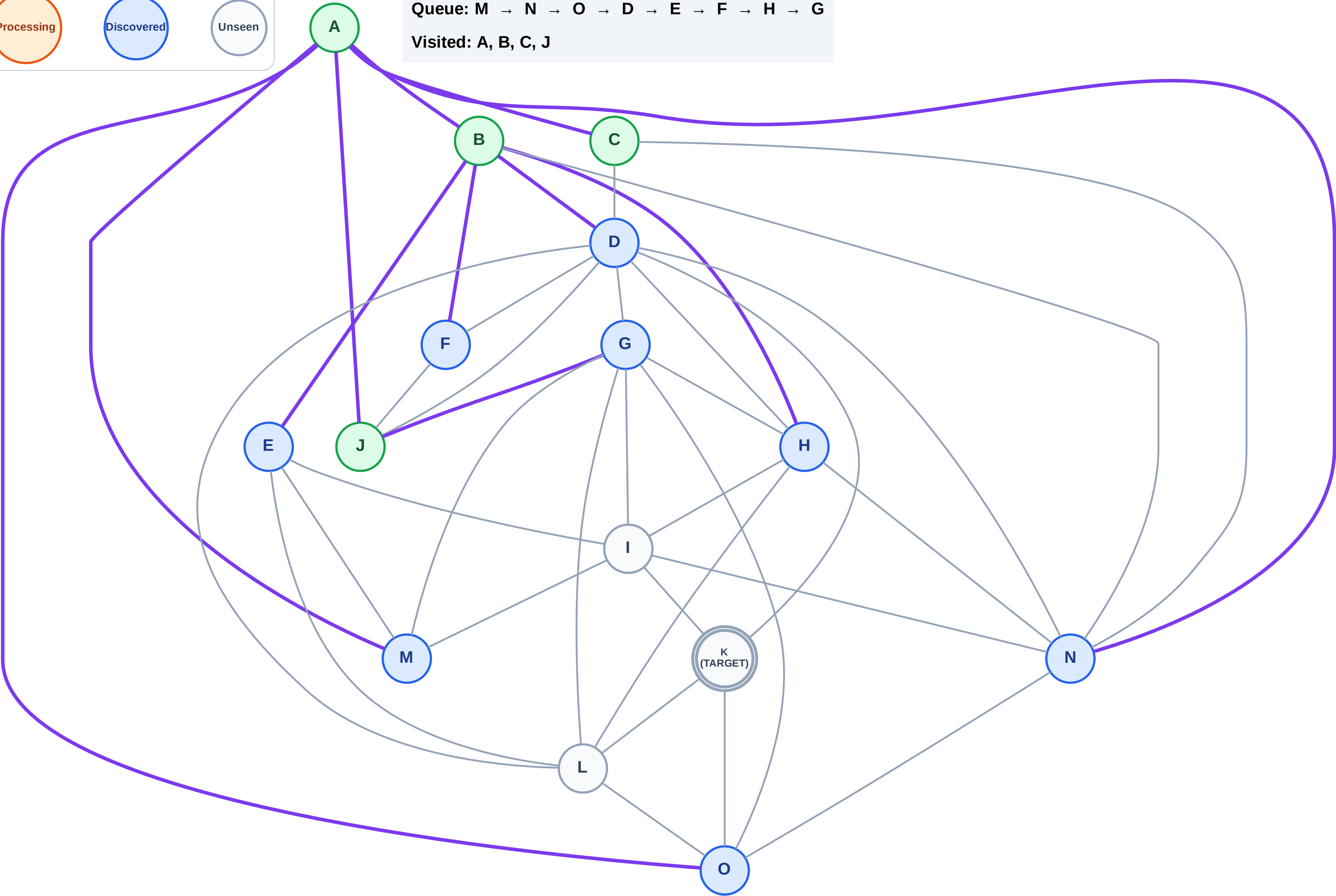
Processing

Discovered

Unseen

Queue: M → N → O → D → E → F → H → G

Visited: A, B, C, J



BFS Traversal

Step 10: Process node M

Legend

Complete

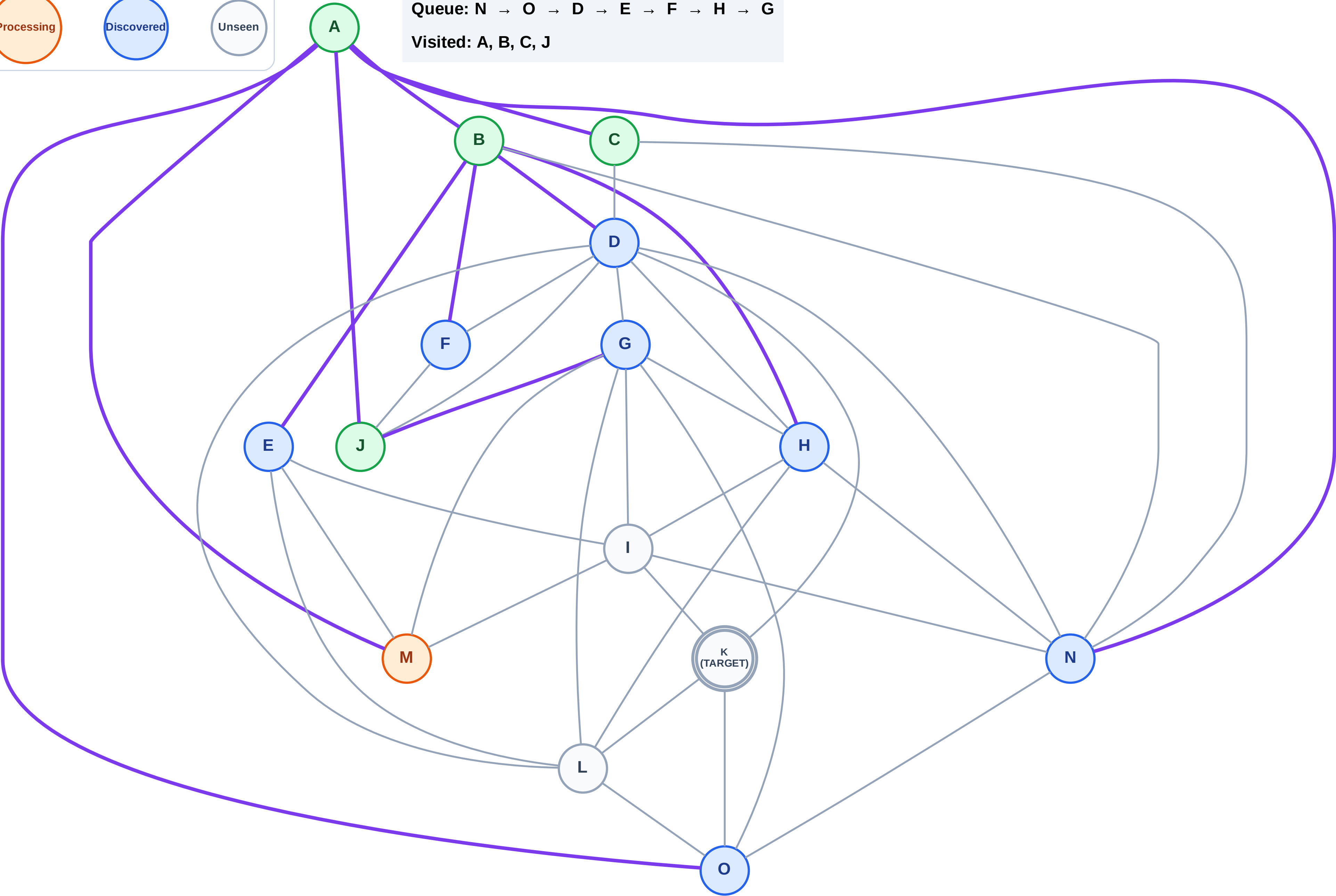
Processing

Discovered

Unseen

Queue: N → O → D → E → F → H → G

Visited: A, B, C, J



Step 11: Discover I from M

Complete



Processing

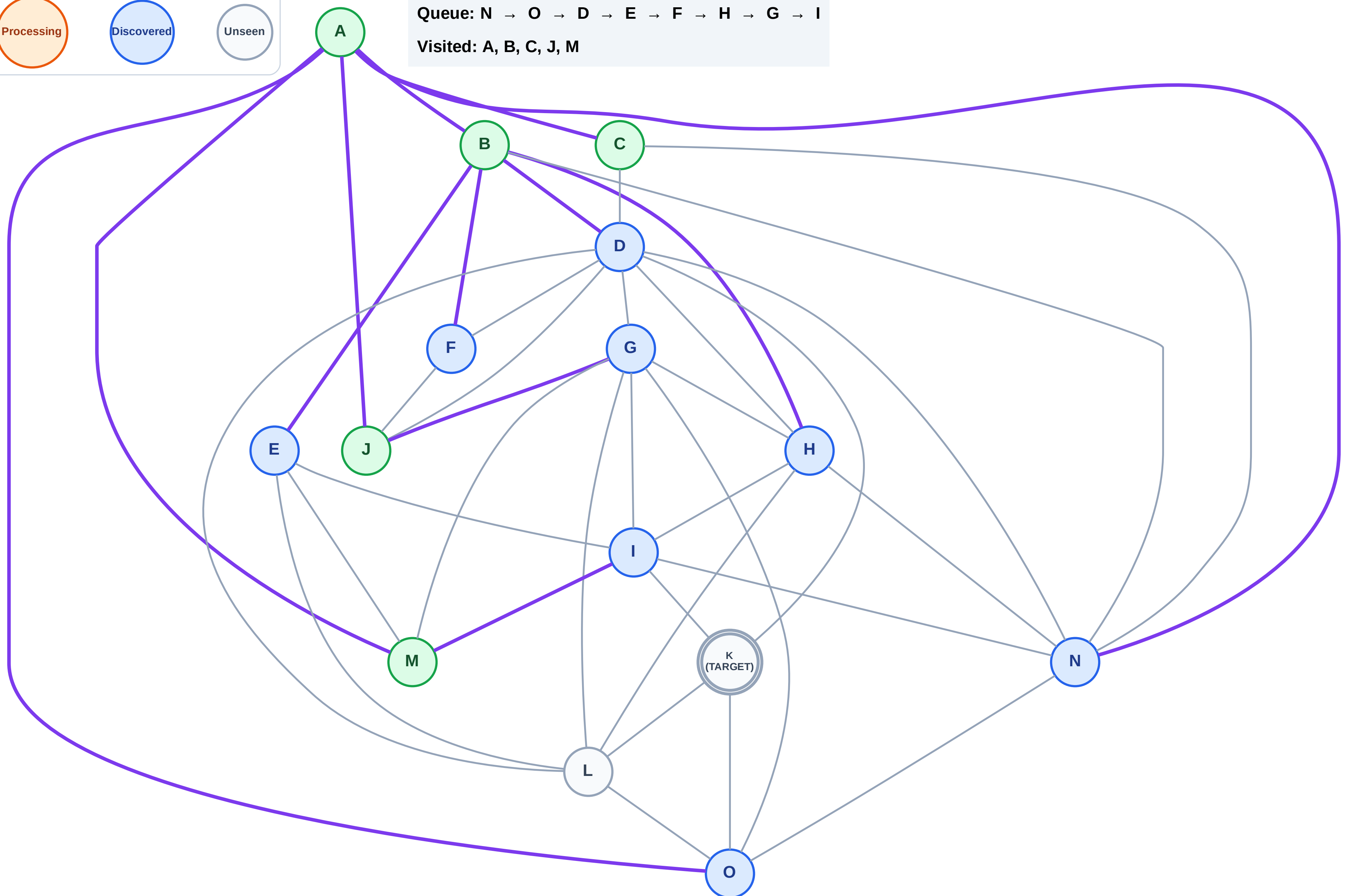


Discovered

Unseen

Unseen

Visited: A, B, C, J, M



BFS Traversal

Step 12: Process node N

Legend

Complete

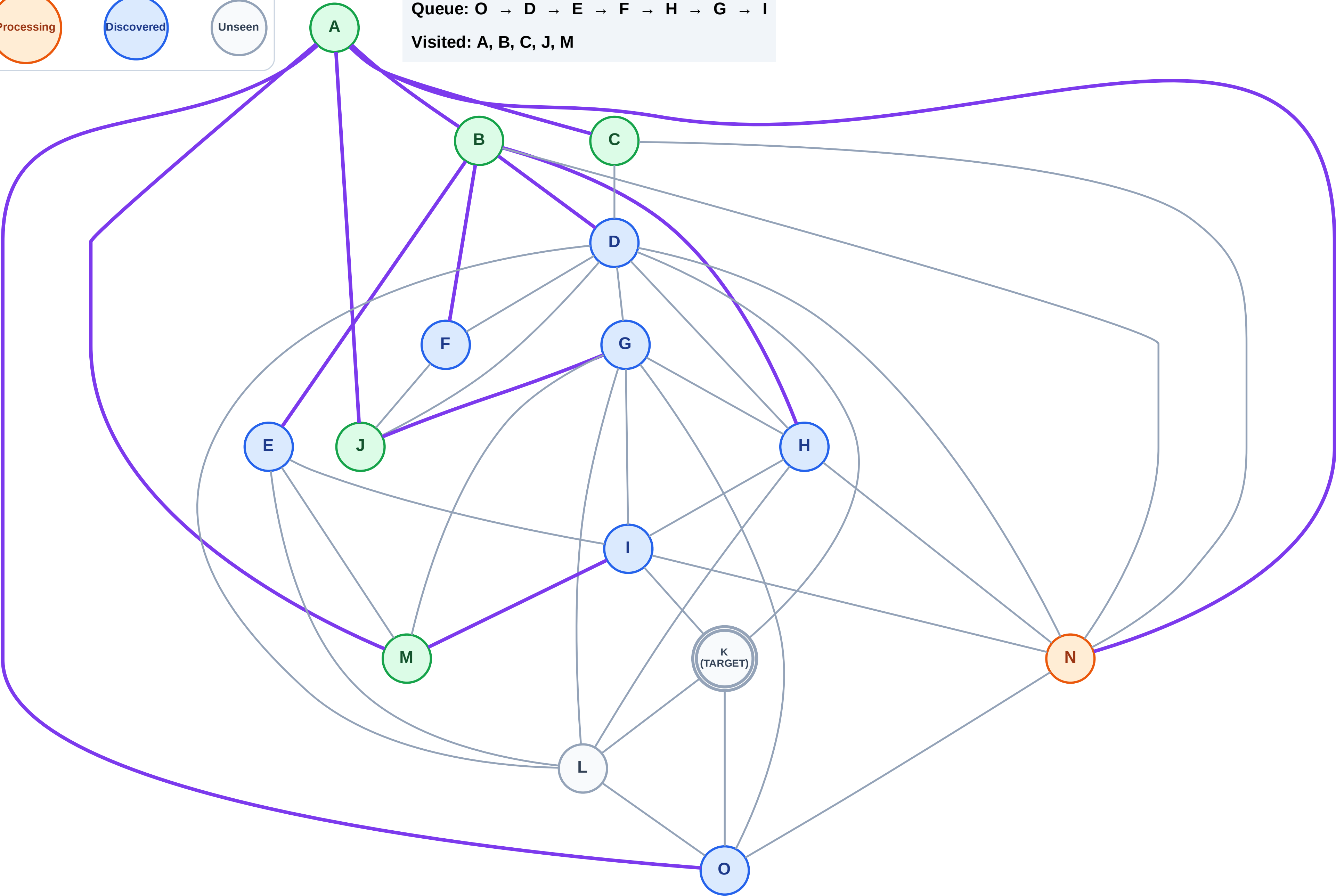
Processing

Discovered

Unseen

Queue: O → D → E → F → H → G → I

Visited: A, B, C, J, M



BFS Traversal

Step 13: No new neighbors from N

Legend

Complete

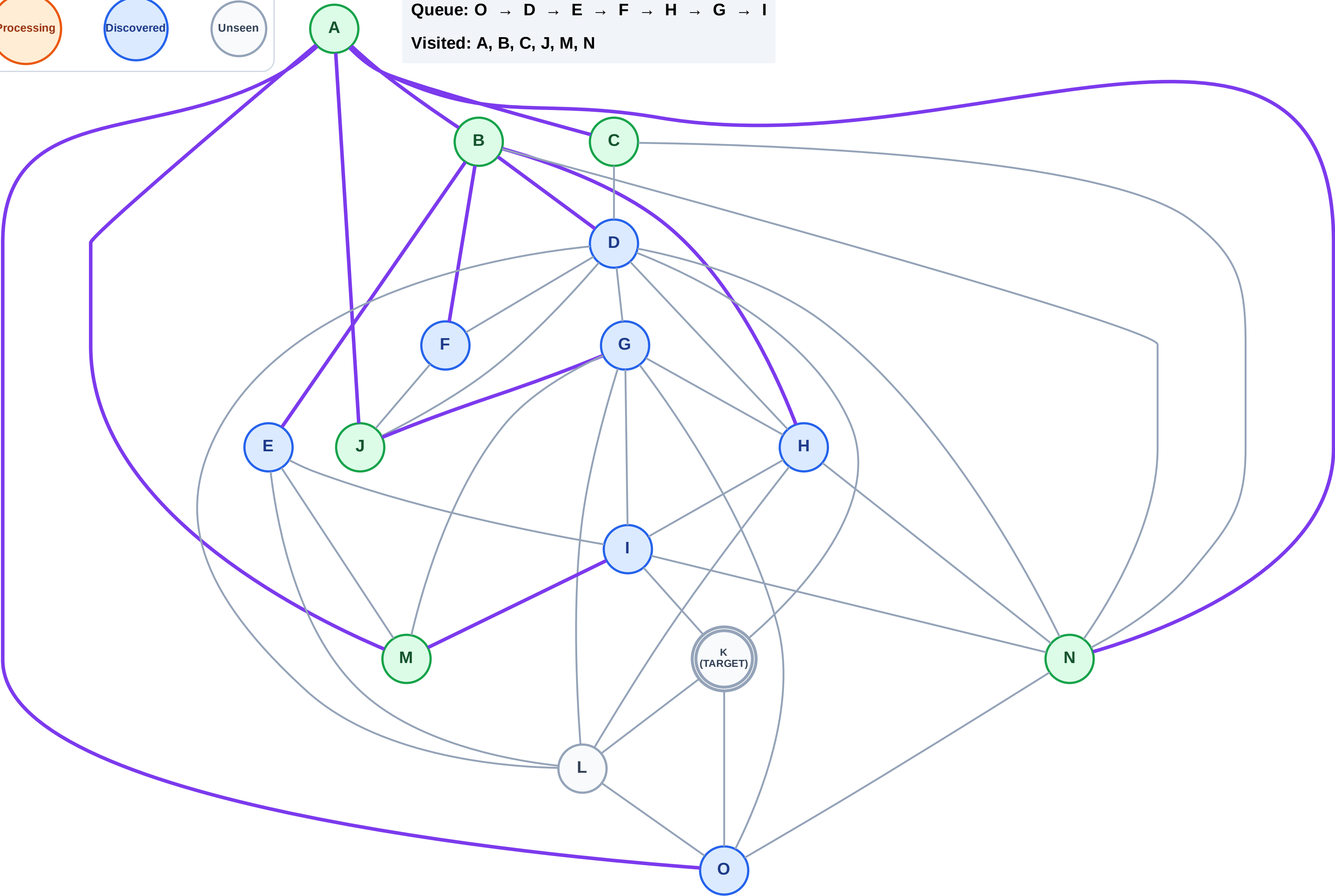
Processing

Discovered

Unseen

Queue: O → D → E → F → H → G → I

Visited: A, B, C, J, M, N



BFS Traversal

Step 14: Process node O

Legend

Complete

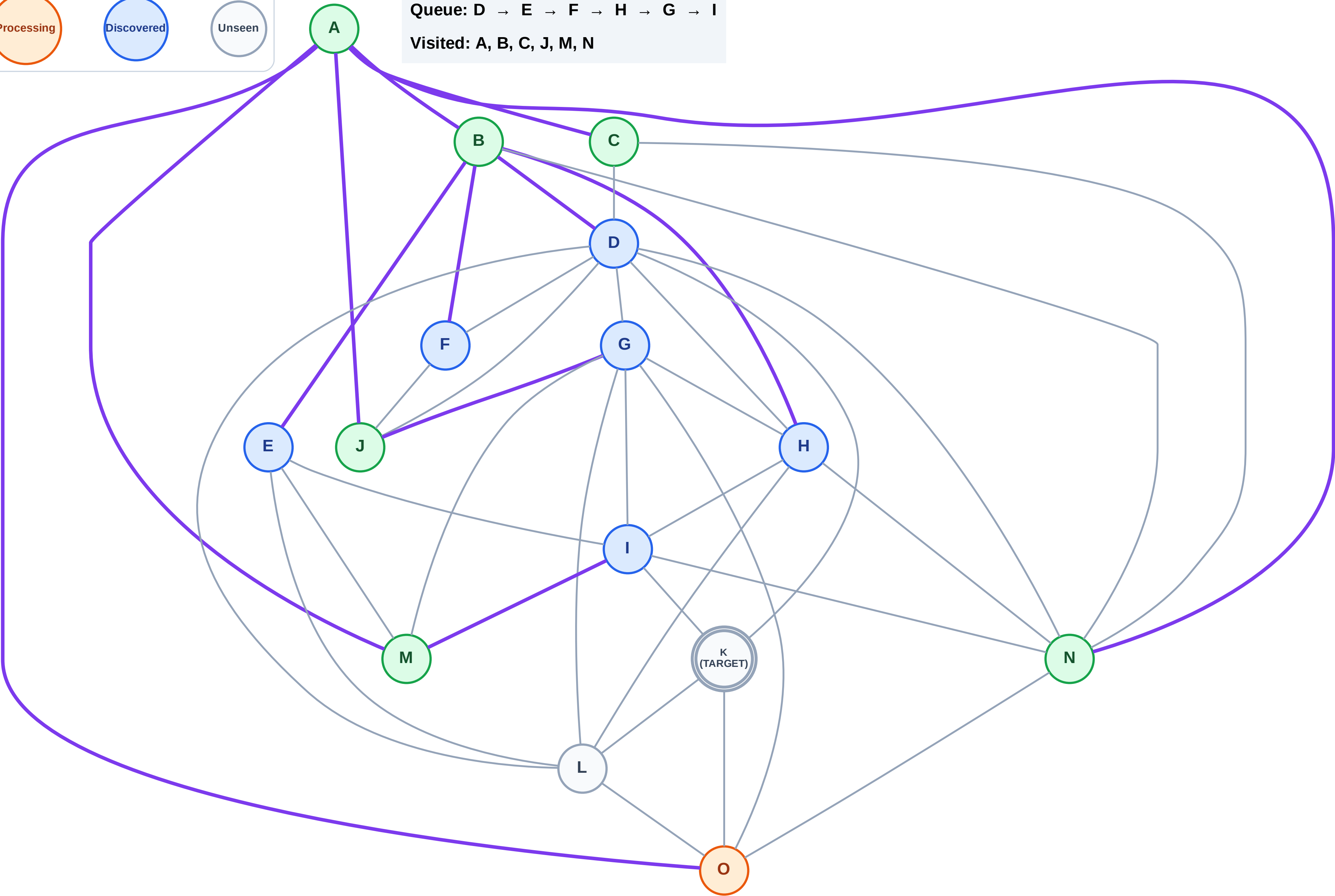
Processing

Discovered

Unseen

Queue: D → E → F → H → G → I

Visited: A, B, C, J, M, N



BFS Traversal

Step 15: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → O → K

Legend

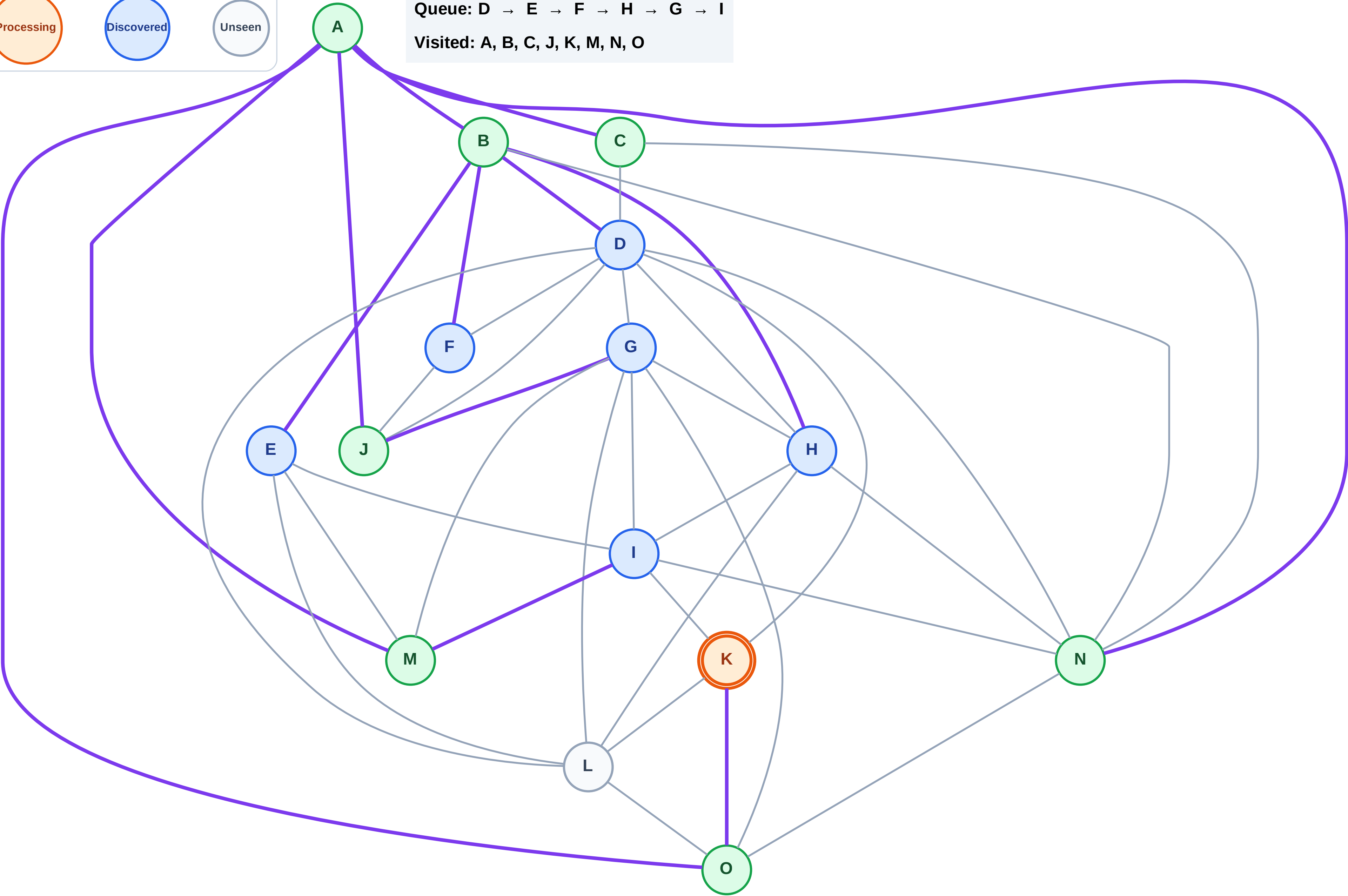
Complete

Processing

Discovered

Unseen

Queue: D → E → F → H → G → I
Visited: A, B, C, J, K, M, N, O



DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

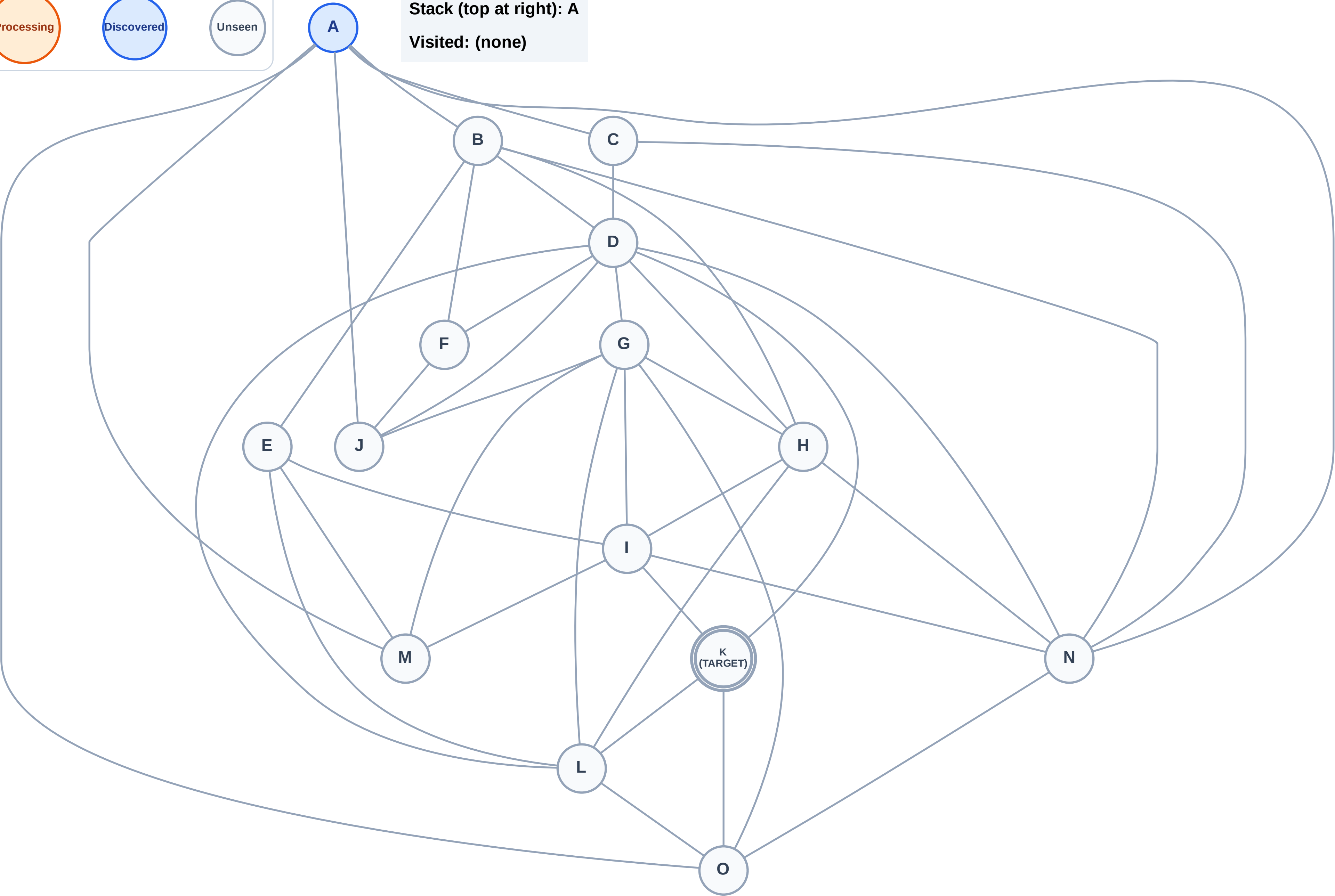
Complete

Processing

Discovered

Unseen

Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

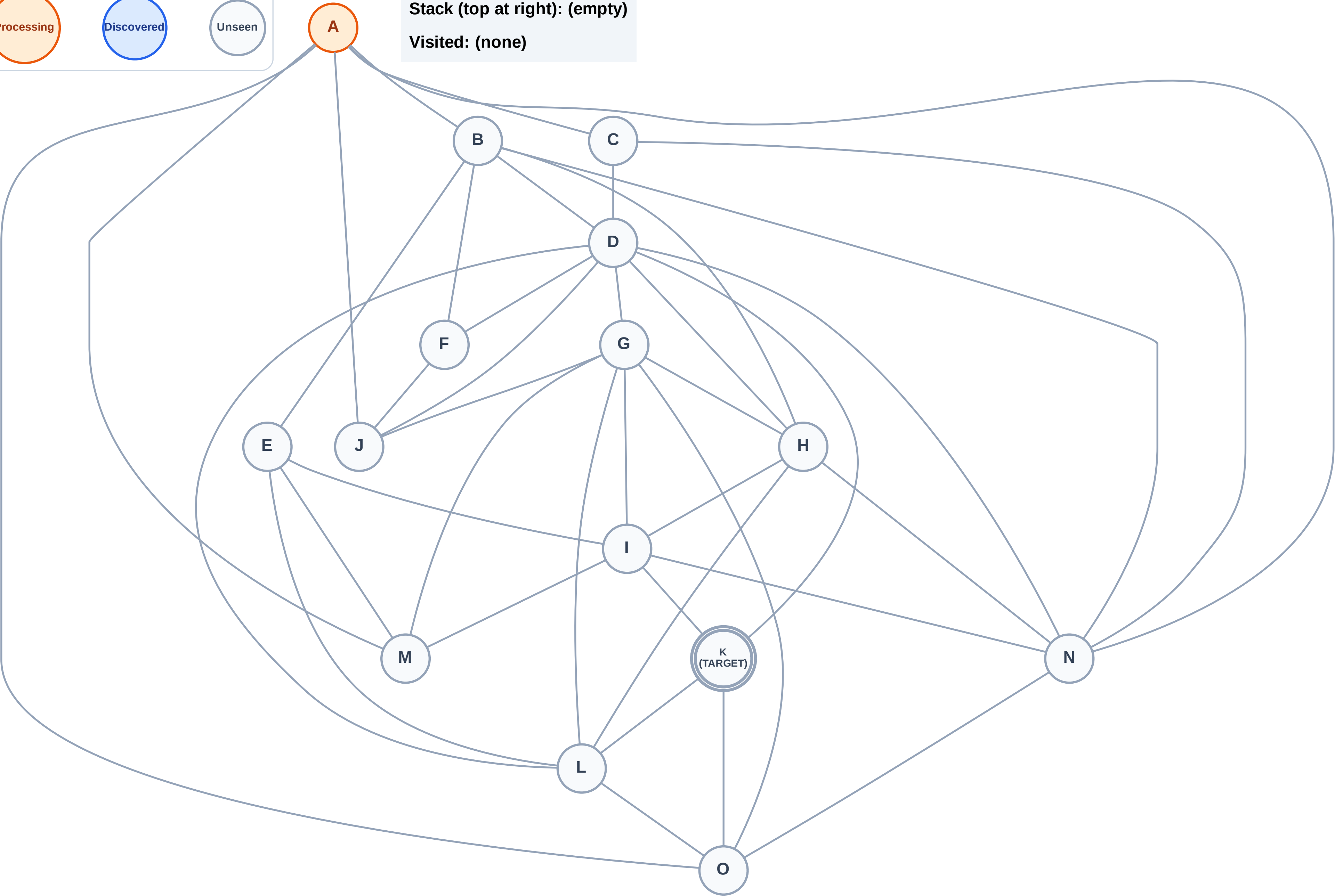
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



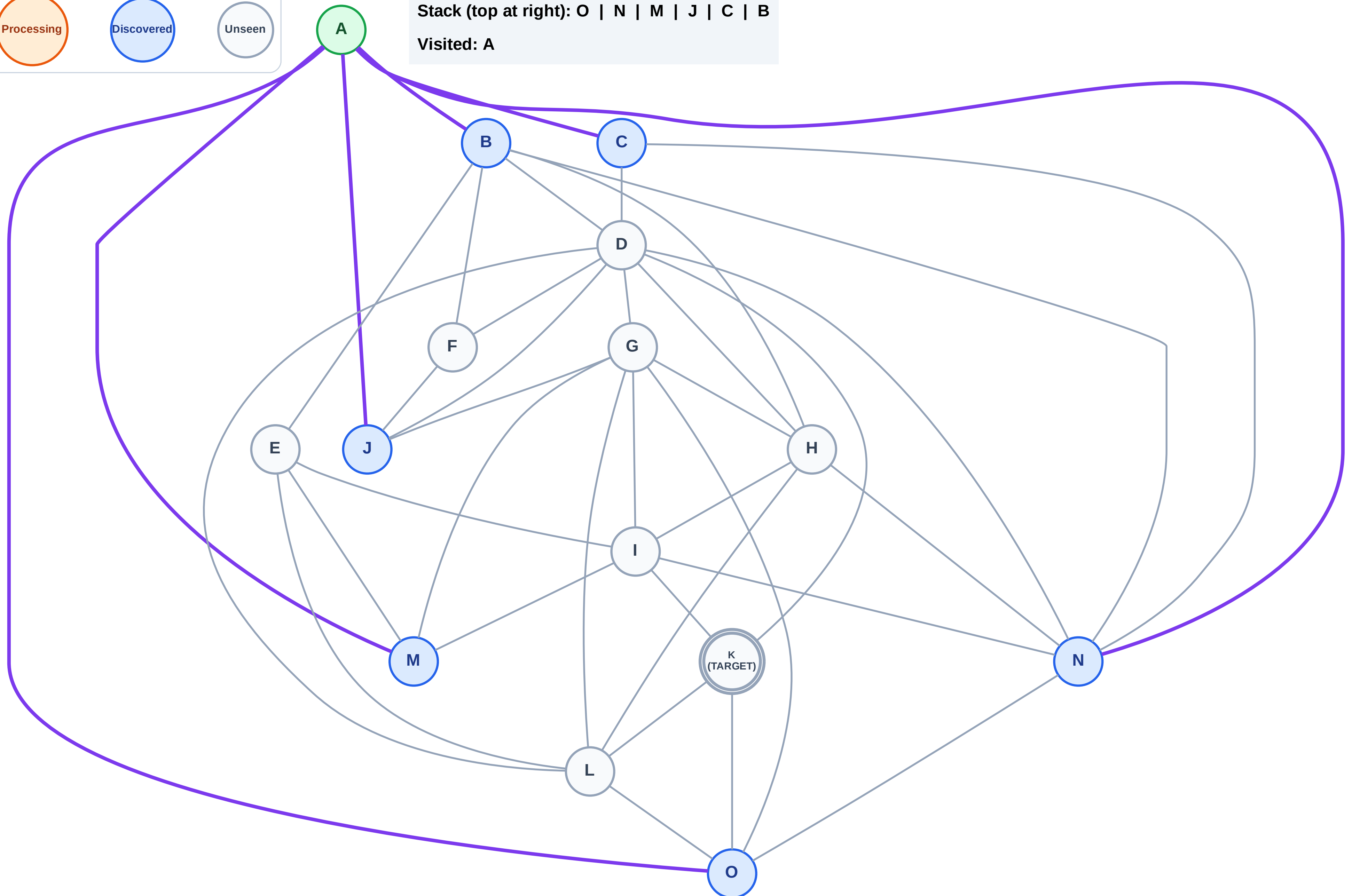
DFS Traversal

Step 3: Discover O, N, M, J, C, B from A

Legend



Stack (top at right): O | N | M | J | C | B
Visited: A



DFS Traversal

Step 4: Process node B

Legend

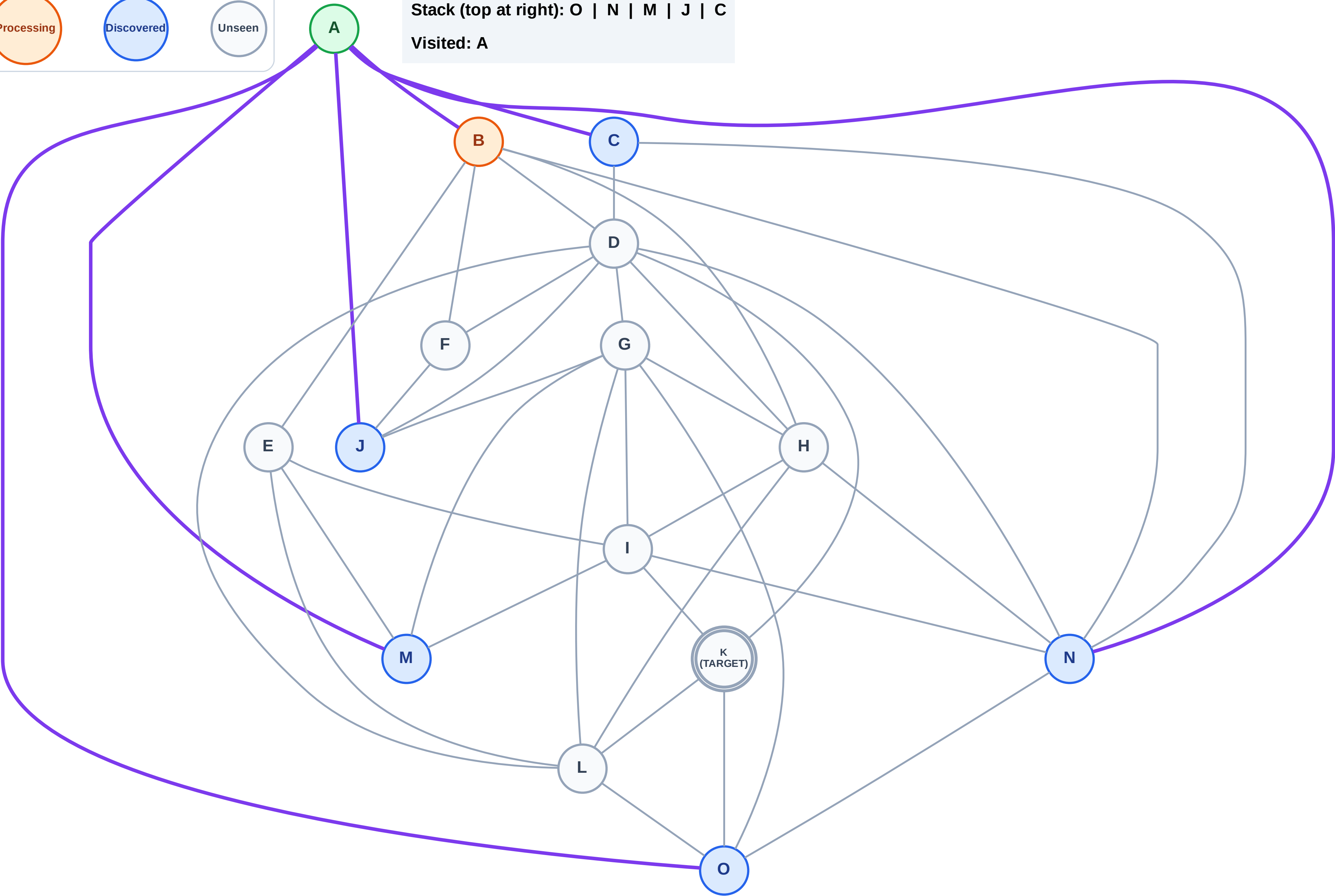
Complete

Processing

Discovered

Unseen

Stack (top at right): O | N | M | J | C
Visited: A



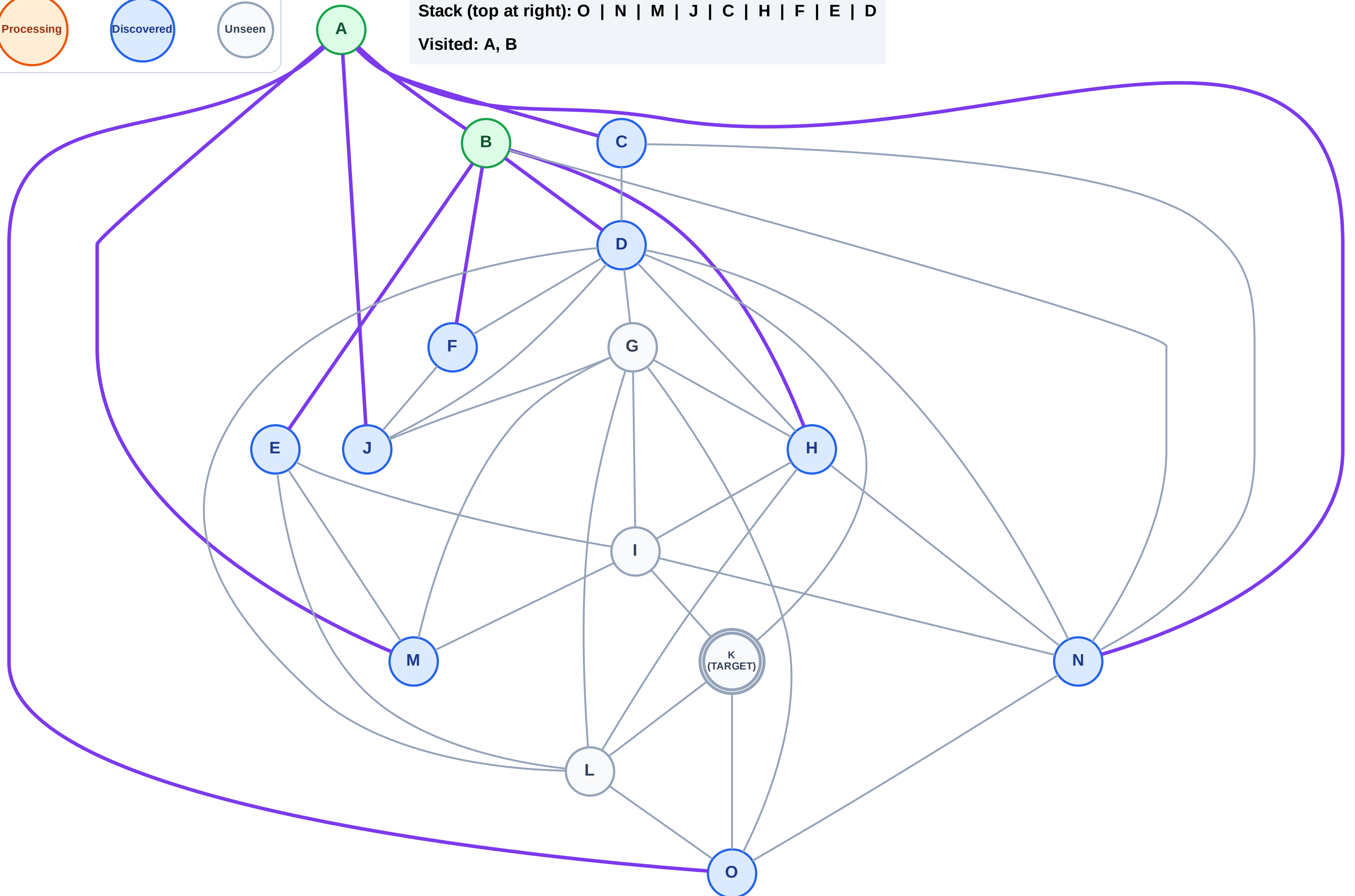
DFS Traversal

Step 5: Discover H, F, E, D from B

Legend



Stack (top at right): O | N | M | J | C | H | F | E | D
Visited: A, B



DFS Traversal

Step 6: Process node D

Legend

Complete

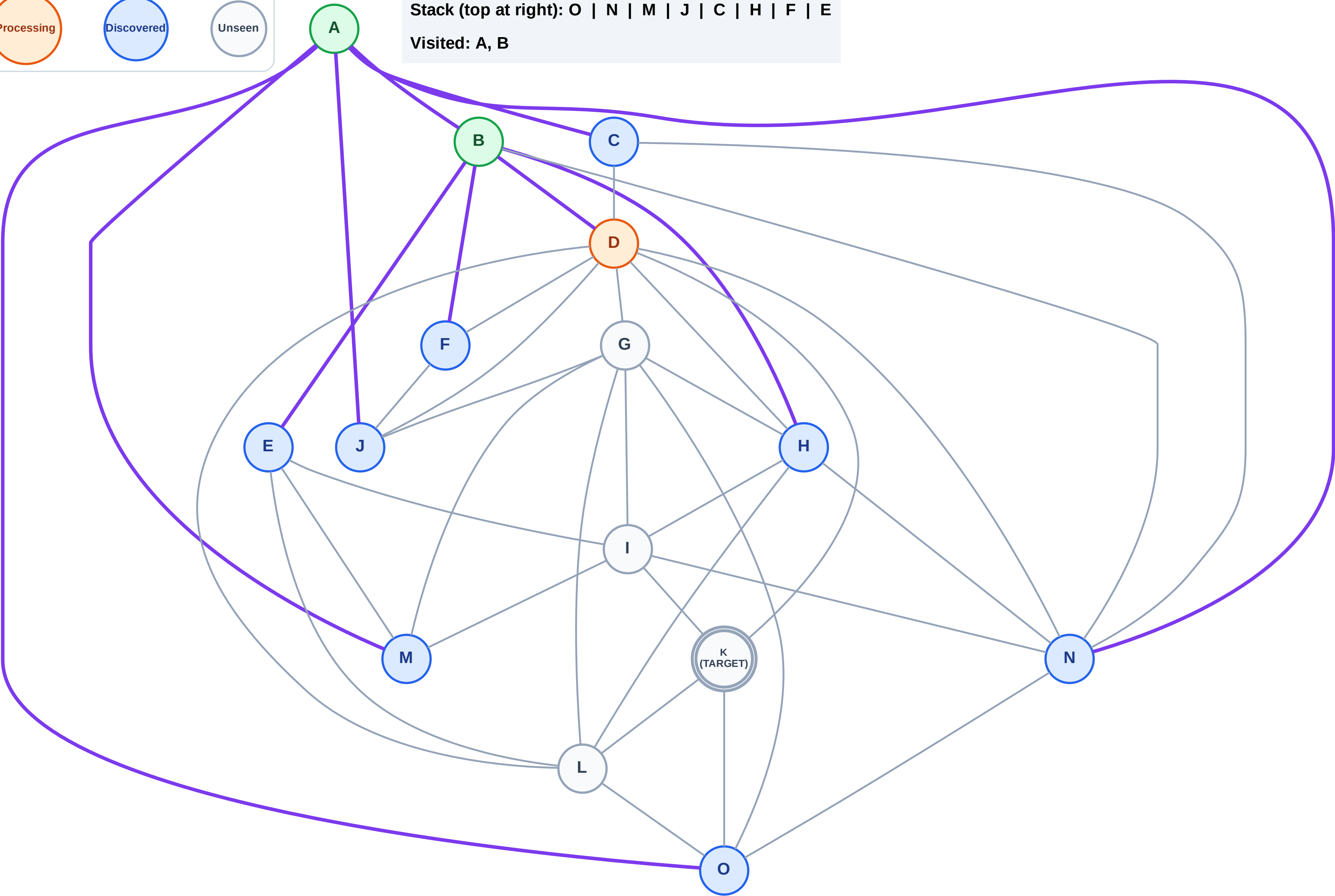
Processing

Discovered

Unseen

Stack (top at right): O | N | M | J | C | H | F | E

Visited: A, B



DFS Traversal

Step 7: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (3 hops): A → B → D → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): O | N | M | J | C | H | F | E | L

Visited: A, B, D, K

